Andrew McElhannon

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BIO

Computer Science graduate (3.9 GPA) with Security+ and Network+ certifications and hands-on experience in vulnerability assessment, firewall configuration, networking, and AI-driven security tooling. Seeking entry-level programming and networking roles.

EDUCATION

Shiley-Marcos School of Engineering, University of San Diego San Diego, CA
B.A. Computer Science May 2026
Final G.P.A: 3.9/4.0
Certifications: **Security+, Network+**

SKILLS/ CERTIFICATIONS

Languages: Python, Java, C, SQL, HTML/CSS, SystemVerilog
Tools/Technologies: Git, GitHub, Linux, REST APIs, Socket Programming, Virtual Machines, pfSense, Nessus, iOS, Windows

UNIVERSITY PROJECTS

Adversys (Senior Project), University of San Diego Fall 2025 - Spring 2026

- Modified the existing codebase to enable AI agents to identify vulnerabilities using Adversys security tooling and support automated remediation-report generation
- Collaborated with a four-person development team to enhance an existing cybersecurity application and support deeper integration of embedded AI agents and security tools
- Designed the integration with an emphasis on safe, secure, and reliable automation of vulnerability assessment and remediation workflows
- Developed within an Agile workflow using two-week sprints, peer reviews, technical presentations, and iterative software development

Streaming Music Protocol, University of San Diego Spring 2026

- Designed a client-server music streaming application using TCP sockets to transmit binary audio and resource requests
- Implemented a custom application-layer protocol and server-side parser supporting request validation and error detection
- Resolved cross-language binary compatibility between Java and C, including byte ordering, memory alignment, and data serialization

Cybersecurity Network, University of San Diego Fall 2025

- Co-architected a network of 7 Linux-based VMs on a physical host to simulate attacks and network routing/management
- Configured and deployed a pfSense firewall as a default gateway to audit and protect network traffic
- Executed vulnerability assessments using Nessus scans and performed brute-force password attacks

Generallt, John Cabot University April 2025

- Designed and deployed a minimum viable prototype e-commerce website using HTML/CSS
- Engineered a responsive frontend layout with intuitive UI, clear navigation, and a modern aesthetic design
- Demonstrated intended functionality and user flows that could be shown to a potential client

SurfBuddy, University of San Diego December 2024

- Co-engineered a scalable terminal application integrating APIs to fetch, parse, and display real-time environmental data
- Applied object-oriented design and the Factory Method pattern to construct location and preference objects while separating object creation from application logic